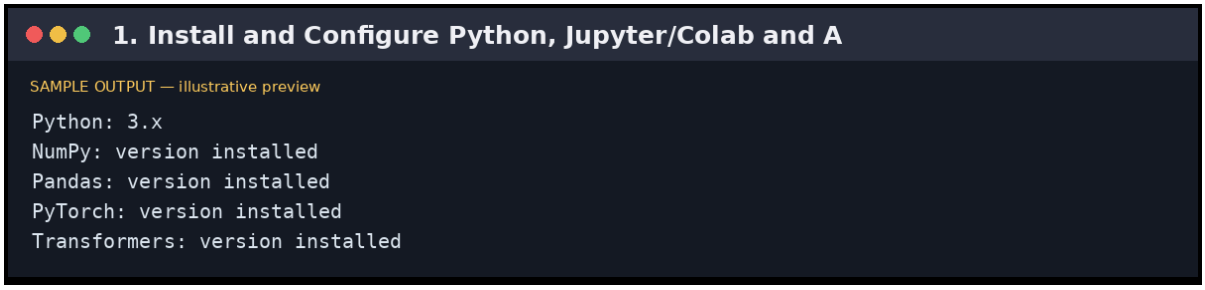
CSA06507 – GENERATIVE AI

1. Install and Configure Python, Jupyter/Colab and AI Libraries

PROGRAM / PROMPT

INPUT

OUTPUT



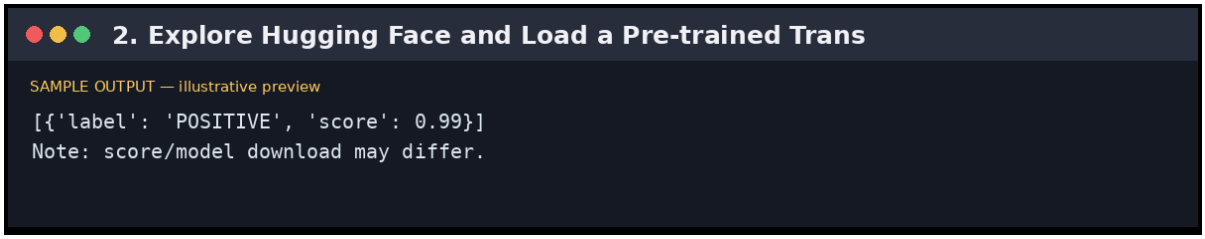
RESULT

2. Explore Hugging Face and Load a Pre-trained Transformer

PROGRAM / PROMPT

INPUT

OUTPUT



RESULT

3. Tokenization Using BERT and GPT Tokenizers

PROGRAM / PROMPT

INPUT

OUTPUT

3. Tokenization Using BERT and GPT Tokenizers

SAMPLE OUTPUT — illustrative preview

```
BERT tokens: ['generative', 'ai', 'is', 'useful', '!']
GPT-2 tokens: token pieces may include leading-space markers
Comparison: token boundaries differ because the tokenizers use different vocabularies
and rules.
```

RESULT

4. Generate Contextual Embeddings Using a Transformer

PROGRAM / PROMPT

[redacted]

INPUT

[redacted]

OUTPUT

4. Generate Contextual Embeddings Using a Transforme

SAMPLE OUTPUT — illustrative preview

```
Input tokens: ['[CLS]', 'ai', 'supports', 'engineering', 'applications', '.', '[SEP]']
Embedding shape: (1, 7, 768)
```

[redacted]

RESULT

[redacted]

5. Text Generation Using a Pre-trained Large Language Model

[redacted]

PROGRAM / PROMPT

[redacted]

INPUT

[redacted]

OUTPUT

5. Text Generation Using a Pre-trained Large Languag

SAMPLE OUTPUT — illustrative preview

```
Artificial intelligence in education ...
The continuation is generated by the model and can vary between runs.
```

[redacted]

RESULT

[redacted]